اقْرَأْ بِاسْمِ رَبِّكَ الَّذِي خَلَقَ

(اے محمدﷺ) اپنے پروردگار کا نام لے کر پڑھو جس نے (سب کچھ) پیدا کیا



VIRTUALIZATION

CS-505 MCQ FINAL

Topics (7-16 Weeks)

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MCQ 1:

What is a Type 2 hypervisor?

A) A hypervisor that runs directly on hardware.

B) A hypervisor that runs on top of an operating system.

C) A hypervisor that runs as an application on an existing OS.

D) A hypervisor used only for cloud computing.

Answer: C

MCQ 2:

Which of the following is an example of a Type 2 hypervisor?

A) VMware ESXi

B) VMware Workstation

C) Microsoft Hyper-V

D) XenServer

Answer: B

MCQ 3:

What is the primary use of VMware Fusion?

A) To run Linux applications on Windows.

B) To run Windows applications on macOS.

C) To manage cloud-based virtual machines.

D) To replace the host operating system.

Answer: B

MCQ 4:

Which version of VMware Workstation is free for personal use?

A) Pro

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B) Player

C) Enterprise

D) Fusion

Answer: B

MCQ 5:

What is a limitation of VMware Player compared to VMware Pro?

A) It cannot run on Windows.

B) It requires a macOS host.

C) It supports only a single guest OS.

D) It lacks graphical interface support.

Answer: C

MCQ 6:

Which operating systems can VirtualBox run on?

A) Only Windows

B) Only Linux

C) Linux, macOS, and Windows

D) Only macOS

Answer: C

MCQ 7:

Where does a Type 2 hypervisor typically run?

A) Directly on server hardware.

B) In a cloud environment.

C) On top of an installed operating system.

D) As a standalone firmware.

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Answer: C

MCQ 8:

What is the main advantage of using a Type 2 hypervisor?

A) Highest performance for enterprise workloads.

B) Ease of use for running multiple OSes on a personal computer.

C) Direct hardware access for low latency.

D) Exclusive support for cloud integration.

Answer: B

MCQ 1:

What is the primary purpose of VMware Workstation 15 Pro?

A) To run directly on hardware without an OS.

B) To create and manage virtual machines on a host OS.

C) To replace the host operating system entirely.

D) To only connect to cloud servers.

Answer: B

MCQ 2:

Which option is available in the VMware Workstation 15 Pro interface?

A) "Delete Host OS"

B) "Create a New Virtual Machine"

C) "Format Physical Disk"

D) "Disable Virtualization"

Answer: B

MCQ 3:

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What is required to install a guest OS in VMware Workstation?

A) A physical CD-ROM drive only.

B) An ISO file or physical installer disc.

C) A pre-installed OS on the host.

D) No installation media is needed.

Answer: B

MCQ 4:

What happens if you select "I will install the OS later" during VM creation?

A) The VM automatically installs Linux.

B) The VM is created with a blank hard disk.

C) The host OS is erased.

D) The VM connects to a cloud server.

Answer: B

MCQ 5:

Where is the default location for storing virtual machines in VMware Workstation?

A) C:\Program Files\VMware\

B) C:\Users\[Username]\Documents\Virtual Machines\

C) D:\VM_Data\

D) The desktop folder.

Answer: B

MCQ 6:

What is the recommended disk size for a Windows 10 VM in this guide?

A) 20 GB

B) 60 GB

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C) 200 GB
D) 500 GB

Answer: B

MCQ 7:

What is an advantage of splitting a virtual disk into multiple files?
A) Improves performance for large disks.
B) Makes it easier to move the VM to another computer.
C) Encrypts the VM automatically.
D) Reduces the total disk space used.

Answer: B

MCQ 8:

Which VMware Workstation feature allows connecting to a remote server?
A) "New Virtual Machine Wizard"
B) "Connect to a Remote Server"
C) "Edit > Preferences"
D) "Help Menu"

Answer: B

MCQ 9:

What type of hypervisor is VMware Workstation 15 Pro?
A) Type 1 (Bare-metal).
B) Type 2 (Hosted).
C) Hybrid hypervisor.
D) Cloud-only hypervisor.

Answer: B

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MCQ 10:

Which option is NOT part of the New Virtual Machine Wizard?

A) Naming the VM.

B) Specifying disk capacity.

C) Formatting the host OS.

D) Selecting guest OS installation media.

Answer: C

MCQ 1:

What type of hypervisor is ESXi?

A) Type 2 (Hosted)

B) Type 1 (Bare-metal)

C) Hybrid

D) Container-based

Answer: B

MCQ 2:

Which of the following is a current version of ESXi mentioned in the document?

A) vSphere 5.5

B) vSphere 6.7

C) vSphere 7.0

D) vSphere 8.0

Answer: B

MCQ 3:

What is the primary role of ESXi?

A) To run as an application on a host OS.

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B) To manage physical resources and assign them to virtual machines.

C) To replace traditional operating systems on personal computers.

D) To only support cloud-based workloads.

Answer: B

MCQ 4:

Which component is part of the vSphere suite?

A) Docker

B) vCenter Server

C) Hyper-V

D) VirtualBox

Answer: B

MCQ 5:

What advantage does ESXi provide in data centers?

A) Increases physical server count.

B) Reduces space, power, and IT administration costs.

C) Requires more hardware resources.

D) Slows down VM performance.

Answer: B

MCQ 6:

Where does ESXi directly run?

A) On top of a Windows/Linux OS.

B) Directly on physical server hardware.

C) Inside a web browser.

D) As a cloud service only.

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Answer: B

MCQ 7:

Which of the following is NOT a physical resource managed by ESXi?

A) CPU

B) Memory

C) Microsoft Office

D) Network Cards

Answer: C

MCQ 8:

What is the relationship between ESXi and vSphere?

A) ESXi is a competitor to vSphere.

B) ESXi is a core component of vSphere.

C) vSphere is a Type 2 hypervisor.

D) They are unrelated technologies.

Answer: B

MCQ 9:

How does ESXi allocate resources to VMs?

A) By requiring manual partitioning for each VM.

B) By dynamically assigning CPU, memory, and storage from the physical server.

C) By relying on the host OS's resource manager.

D) Only via cloud APIs.

Answer: B

MCQ 10:

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Which layer sits between the physical server and virtual machines in ESXi?

A) Host OS (e.g., Windows).

B) Hypervisor (ESXi).

C) Container engine.

D) BIOS firmware.

Answer: B

MCQ 1:

What is the first step to install ESXi 6.5?

A) Purchase a license before downloading.

B) Download the ESXi 6.5 image from VMware's website.

C) Install a host OS like Windows first.

D) Configure a virtual machine in VirtualBox.

Answer: B

MCQ 2:

How should you boot the system to begin ESXi installation?

A) From an existing OS like Windows.

B) Using a bootable ESXi image (USB/DVD).

C) Via a cloud-based installer.

D) Through a network PXE boot only.

Answer: B

MCQ 3:

What must you check before installing ESXi to ensure hardware compatibility?

A) Windows Update history.

B) VMware's Compatibility Guide.

C) BIOS manufacturer's website.

D) CPU temperature logs.

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Answer: B

MCQ 4:

What action is required during the End User License Agreement (EULA) step?

A) Contact VMware support.

B) Accept the EULA to proceed.

C) Skip the EULA and install anyway.

D) Download additional license files.

Answer: B

MCQ 5:

Which disk configuration is supported during ESXi installation?

A) RAID 0 with 4 disks minimum.

B) A single disk with a VMFS partition.

C) Only external USB drives.

D) Cloud storage buckets.

Answer: B

MCQ 6:

What is the default evaluation period for ESXi after installation?

A) 7 days.

B) 60 days.

C) 1 year.

D) Unlimited.

Answer: B

MCQ 7:

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Which step follows setting the root password?

A) Configuring a guest OS.

B) Confirming the installation (disk wipe warning).

C) Joining a Active Directory domain.

D) Installing vCenter Server.

Answer: B

MCQ 8:

What must you do after ESXi installation completes?

A) Keep the ISO mounted for updates.

B) Remove the installation media before rebooting.

C) Install a Type 2 hypervisor on top.

D) Format the disk again.

Answer: B

MCQ 9:

How can you administer ESXi after installation?

A) Only via SSH.

B) Using a web browser or the Direct Console UI.

C) Through Windows Control Panel.

D) By reinstalling the host OS.

Answer: B

MCQ 10:

What happens if you don't register a license after the evaluation period?

A) ESXi stops working entirely.

B) ESXi continues to run but requires a license for full features.

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C) The system reverts to a Type 2 hypervisor.
D) VMware automatically sends a free license.

Answer: B

MCQ 11:

During installation, what does the warning "This disk will be repartitioned" mean?
A) Data on adjacent disks will be erased.
B) All existing data on the selected disk will be lost.
C) The disk will be encrypted automatically.
D) The disk will be converted to FAT32.

Answer: B

MCQ 12:

Which keyboard layout is commonly selected for English users during installation?
A) French AZERTY.
B) US Default (QWERTY).
C) German QWERTZ.
D) Japanese.

Answer: B

MCQ 1:

What is the first step to configure an ESXi host after installation?
A) Create a virtual machine.
B) Authenticate with the root username and password.
C) Install vCenter Server.
D) Configure a guest OS.

Answer: B

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MCQ 2:

Which option is used to secure ESXi host access?

A) Disabling the management network.

B) Configuring a strong root password.

C) Skipping the EULA.

D) Using default credentials.

Answer: B

MCQ 3:

What does "Configure Lockdown Mode" do in ESXi?

A) Allows unrestricted root access.

B) Restricts direct root access to enhance security.

C) Disables all virtual machines.

D) Enables automatic updates.

Answer: B

MCQ 4:

How is the management network typically configured in ESXi?

A) Only via DHCP.

B) Either static IP or DHCP, depending on network requirements.

C) Cloud-based IP assignment.

D) Through a guest OS.

Answer: B

MCQ 5:

What is the purpose of the "Test Management Network" feature?

A) To create a new VLAN.

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B) To verify connectivity (e.g., ping the gateway).

C) To reset the root password.

D) To install additional drivers.

Answer: B

MCQ 6:

Which action is part of troubleshooting ESXi connectivity issues?

A) Deleting virtual machines.

B) Restarting the management network.

C) Reinstalling ESXi.

D) Disabling IPv4.

Answer: B

MCQ 7:

What happens when you "Reset System Configuration" in ESXi?

A) Only the root password is retained.

B) All settings revert to defaults, and VMs are unregistered.

C) The host OS is reinstalled.

D) The hardware is formatted.

Answer: B

MCQ 8:

Where can you view system logs in ESXi for troubleshooting?

A) Windows Event Viewer.

B) ESXi Direct Console UI or web client.

C) VMware Fusion.

D) BIOS settings.

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Answer: B

MCQ 9:

What is the purpose of the "View Support Information" option?

A) To download new ISO images.

B) To access serial numbers, licenses, and VMware support links.

C) To configure firewalls.

D) To enable SSH.

Answer: B

MCQ 10:

Which keyboard layout is commonly selected for English-speaking users?

A) French AZERTY.

B) US Default (QWERTY).

C) German QWERTZ.

D) Japanese.

Answer: B

MCQ 11:

What is a critical warning before resetting ESXi system configuration?

A) It upgrades the ESXi version.

B) Running VMs will be unregistered, and the root password is cleared.

C) It enables Lockdown Mode permanently.

D) It formats all physical disks.

Answer: B

MCQ 12:

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Which feature helps balance outgoing traffic and provides fault tolerance?

A) Single network adapter.

B) Multiple network adapters (load balancing).

C) Disabling IPv6.

D) Using a wireless connection.

Answer: B

MCQ 1:

What is the first step in creating a virtual machine in ESXi?

A) Configuring the network adapter.

B) Creating a datastore or ensuring one exists.

C) Installing the guest OS directly.

D) Setting up vCenter Server.

Answer: B

MCQ 2:

Which disk type is recommended for a new virtual disk in ESXi?

A) IDE

B) SCSI

C) NVMe (for powered-off VMs only)

D) USB

Answer: B

MCQ 3:

What is the purpose of the "datastore browser" in ESXi?

A) To configure CPU settings.

B) To upload ISO images or manage files on a datastore.

C) To enable SSH access.

D) To reset the root password.

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Answer: B

MCQ 4:

How can you allocate disk space for a virtual machine?

A) Only as a single preallocated file.

B) As a single file or split into multiple files, with optional preallocation.

C) Exclusively via cloud storage.

D) Only during VM power-off.

Answer: B

MCQ 5:

What is required to install a guest OS in a new VM?

A) A physical DVD drive.

B) An ISO file uploaded to a datastore or connected via CD/DVD.

C) A preinstalled OS template.

D) DHCP-enabled networking.

Answer: B

MCQ 6:

Which option allows deploying a VM from a preconfigured template?

A) "Create a new virtual machine."

B) "Deploy from an OVF/OVA file."

C) "Clone an existing VM."

D) "Import from vCenter."

Answer: B

MCQ 7:

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What is the default compatibility setting for a new VM in ESXi 7.0?

A) ESXi 5.5.

B) ESXi 7.0 U1.

C) VMware Workstation.

D) Hyper-V.

Answer: B

MCQ 8:

Which feature is enabled by selecting "Allocate all disk space now"?

A) Automatic guest OS updates.

B) Improved performance (but requires immediate disk space).

C) Network load balancing.

D) Fault tolerance.

Answer: B

MCQ 9:

How do you attach an ISO to a VM during creation?

A) Via USB drive.

B) By selecting "Datastore ISO file" for the CD/DVD drive.

C) Using a physical disk partition.

D) Through BIOS settings.

Answer: B

MCQ 10:

What happens if you skip OS installation during VM creation?

A) The VM boots into a recovery console.

B) The VM is created with a blank disk (OS can be installed later).

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C) The host reboots automatically.

D) The datastore is erased.

Answer: B

MCQ 11:

Which tab is used to configure VM hardware settings (e.g., CPU, memory)?

A) "Monitor."

B) "Customize settings."

C) "Datastore browser."

D) "Networking."

Answer: B

MCQ 12:

What is the purpose of the "VMFS" datastore type?

A) To host physical disks only.

B) To store VM files (disks, ISOs, etc.) on ESXi.

C) To enable cloud backups.

D) To replace the host OS.

Answer: B

MCQ 13:

Which action is performed in the "Virtual Machines" tab?

A) Configuring host firewall rules.

B) Creating, registering, or powering on VMs.

C) Upgrading ESXi firmware.

D) Resetting system configuration.

Answer: B

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MCQ 14:

What is the maximum character limit for a VM name in ESXi?

A) 50.

B) 80.

C) 120.

D) 255.

Answer: B

MCQ 15:

Which option ensures a CD/DVD drive is connected at VM power-on?

A) "Enable USB passthrough."

B) "Connect at power on" checkbox.

C) "Allocate all disk space."

D) "Enable fault tolerance."

Answer: B

MCQ 1:

What type of virtualization product is Oracle VM VirtualBox?

A) A cloud-only virtualization platform.

B) A cross-platform virtualization product.

C) A proprietary enterprise solution.

D) A hardware-based hypervisor.

Answer: B

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MCQ 2:

Which computing architectures does VirtualBox support?

A) ARM only.

B) x86 and AMD64/Intel64.

C) RISC-V exclusively.

D) Mainframe architectures.

Answer: B

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MCQ 3:

Under which license is VirtualBox distributed?

A) MIT License.

B) GNU General Public License (GPL).

C) Apache License 2.0.

D) BSD License.

Answer: B

MCQ 4:

On which operating systems can VirtualBox run?

A) Windows only.

B) Windows, Linux, Mac, and Solaris.

C) Linux only.

D) Cloud platforms only.

Answer: B

MCQ 5:

What interfaces does VirtualBox provide for user interaction?

A) Only a graphical user interface (GUI).

B) Both graphical (GUI) and command-line (CLI) interfaces.

C) Only a web-based interface.

D) A mobile app interface.

Answer: B

MCQ 6:

Which of the following is a use case for VirtualBox?

A) Running a guest OS like Windows 7 on a host OS like Windows 10.

B) Managing large-scale data centers.

C) Replacing the host OS entirely.

D) Only virtualizing cloud servers.

Answer: A

MCQ 7:

What types of virtualization does VirtualBox support?

A) Only software virtualization.

B) Both hardware and software virtualization.

C) Only hardware virtualization.

D) Neither hardware nor software virtualization.

Answer: B

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MCQ 8:

For which type of projects is VirtualBox recommended?

A) Large-scale enterprise projects.

B) Small and mid-range projects.

C) High-performance computing clusters.

D) Cloud-native applications only.

Answer: B

MCQ 9:

What is required before installing guest OSes in VirtualBox?

A) A cloud subscription.

B) Installing VirtualBox on the host OS.

C) Purchasing a proprietary license.

D) Disabling the host OS.

Answer: B

MCQ 10:

Which of the following is true about VirtualBox?

A) It is a Type 1 hypervisor.

B) It is a Type 2 hypervisor.

C) It only runs on Linux hosts.

D) It requires dedicated server hardware.

Answer: B

MCQ 1:

What type of hypervisor is VMware ESXi?

A) Type 2

B) Type 1

C) Hybrid

D) Container-based

Answer: B

MCQ 2:

Which virtualization platform is open-source?

A) VMware Workstation

B) VirtualBox

C) VMware Fusion

D) Hyper-V

Answer: B

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MCQ 3:

What virtualization types does VirtualBox support?

A) Only hardware virtualization

B) Both hardware and software virtualization

C) Only software virtualization

D) Neither

Answer: B

MCQ 4:

Which platform requires a paid license for advanced features?

A) VMware (except Workstation Player)

B) VirtualBox

C) Both

D) Neither

Answer: A

MCQ 5:

Which disk format is supported by both VMware and VirtualBox?

A) VHD

B) VMDK

C) QCOW2

D) RAW

Answer: B

MCQ 6:

Which platform is recommended for enterprise projects?

A) VMware

B) VirtualBox

C) Both equally

D) Hyper-V

Answer: A

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MCQ 7:

Which feature is free in VirtualBox but requires payment in VMware?

A) Hardware virtualization

B) Snapshots

C) USB support

D) Network bridging

Answer: B

Microsoft

MCQ 8:

Which operating systems can run VirtualBox?

A) Windows only

B) Windows, Linux, macOS, Solaris

C) Linux only

D) Cloud platforms only

Answer: B

MCQ 9:

Which platform generally offers better VM performance?

A) VMware

B) VirtualBox

C) Both are equal

D) Depends on host OS

Answer: A

MCQ 10:

What is needed to run VMware on macOS?

A) Workstation Player

B) VMware Fusion/Fusion Pro

C) VirtualBox

D) Boot Camp

Answer: B

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MCQ 11:

Which platform is better for small to mid-level projects?

A) VMware

B) VirtualBox

C) Both are equal

D) Neither

Answer: B

MCQ 12:

What is a key advantage of VirtualBox over VMware?

A) Better enterprise support

B) Free snapshots

C) Faster VM performance

D) Only Type-1 hypervisor

Answer: B

MCQ 1:

What type of hypervisor is Microsoft Hyper-V?

A) Type 2

B) Type 1

C) Hybrid

D) Container-based

Answer: B

MCQ 2:

Which company developed Hyper-V?

A) Oracle

B) Microsoft

C) VMware

D) IBM

Answer: B

MCQ 3:

How can Hyper-V be enabled?

A) By purchasing a separate license

B) Through the Windows operating system

C) Only on cloud platforms

D) Via third-party software

Answer: B

MCQ 4:

Which of the following is an advantage of Hyper-V?

A) Increased power consumption

B) Reduced hardware costs

C) Limited flexibility

D) Single-environment support

Answer: B

MCQ 5:

What feature does Hyper-V use for workload migration?

A) vMotion

B) Live Migration

C) Fault Tolerance

D) Snapshots

Answer: B

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MCQ 6:

Which operating systems does Hyper-V support as guests?

A) Windows only

B) Windows and Linux

C) macOS only

D) Solaris only

Answer: B

MCQ 7:

What is the primary purpose of Hyper-V?

A) To replace physical servers entirely

B) To create multiple isolated environments on a single server

C) To manage cloud storage

D) To enhance graphics performance

Answer: B

MCQ 8:

Which term does Hyper-V use for its snapshot feature?

A) Checkpoints

B) Snapshots

C) Backups

D) Clones

Answer: A

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MCQ 9:

How does Hyper-V contribute to power efficiency?

A) By increasing hardware usage

B) By reducing the need for multiple physical servers

C) By disabling virtualization

D) By limiting VM creation

Answer: B

MCQ 10:

Which tool is equivalent to VMware's vMotion in Hyper-V?

A) Fault Tolerance

B) Live Migration

C) Checkpoints

D) Disaster Recovery

Answer: B

Microsoft

MCQ 11:

What is a key market competitor to Hyper-V?

A) VirtualBox

B) VMware

C) Docker

D) Kubernetes

Answer: B

MCQ 12:

Which of the following is NOT true about Hyper-V?

A) It is a Type 1 hypervisor.

B) It supports Linux guest OSes.

C) It is an open-source platform.

D) It reduces hardware costs.

Answer: C

MCQ 1:

What is the foundational layer in Hyper-V architecture?

A) Parent Partition

B) Hypervisor

C) Child Partition

D) VMBus

Answer: B

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MCQ 2:

Which component manages hardware resources in Hyper-V?

A) Child Partition

B) Hypervisor

C) VMBus

D) VSC

Answer: B

MCQ 3:

What is the role of the Parent Partition in Hyper-V?

A) Runs guest operating systems

B) Manages virtual machines and provides virtualization services

C) Directly accesses hardware

D) Acts as a backup partition

Answer: B

Microsoft®

MCQ 4:

Which component facilitates communication between partitions in Hyper-V?

A) VSC

B) VMBus

C) Hypervisor

D) Hardware

Answer: B

MCQ 5:

What does VSC stand for in Hyper-V architecture?

A) Virtual System Controller

B) Virtualization Service Client

C) Virtual Server Component

D) Virtual Storage Controller

Answer: B

MCQ 6:

Which partition hosts the guest operating systems in Hyper-V?

A) Parent Partition

B) Child Partition

C) Hypervisor

D) VMBus

Answer: B

MCQ 7:

What is the function of VSP in Hyper-V?

A) Manages hardware directly

B) Provides virtualization services to Child Partitions

C) Acts as a backup service

D) Controls network traffic

Answer: B

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MCQ 8:

Which of the following is NOT a part of Hyper-V architecture?

A) VMBus

B) Docker Container

C) Hypervisor

D) Parent Partition

Answer: B

Microsoft

MCQ 9:

How do Child Partitions access hardware resources?

A) Directly

B) Through the Parent Partition via VMBus

C) Via cloud services

D) Using external drivers

Answer: B

MCQ 10:

Which technology enables high-speed communication between partitions?

A) Ethernet

B) VMBus

C) PCI Express

D) USB

Answer: B

MCQ 11:

What is the primary advantage of Hyper-V's architecture?

A) Limited scalability

B) Efficient resource isolation and management

C) Dependency on physical hardware drivers

D) Single-partition operation

Answer: B

Systems Administrator

MCQ 12:

Which component is responsible for creating and managing virtual machines?

A) Child Partition

B) Parent Partition

C) VMBus

D) Hardware

Answer: B

MCQ 1:

What is the first step to enable Hyper-V on Windows 10 Pro?

A) Install third-party software

B) Enable it through "Turn Windows features on or off"

C) Download it from Microsoft Store

D) Purchase a separate license

Answer: B

Microsoft®

MCQ 2:

Where can you access Hyper-V after enabling it?

A) Control Panel

B) Search bar in Windows

C) Task Manager

D) Device Manager

Answer: B

MCQ 3:

How do you create a new virtual machine in Hyper-V?

A) Right-click on the desktop

B) Use the "New" option in Hyper-V Manager

C) Command Prompt

D) Windows Settings

Answer: B

MCQ 4:

What is the first action to install an OS in a Hyper-V VM?

A) Format the hard disk

B) Connect to the VM and click "Start"

C) Enable BIOS settings

D) Download ISO from the internet

Answer: B

MCQ 5:

Which of the following can you modify in VM settings?

A) Screen resolution

B) Memory and CPU allocation

C) Host OS version

D) Hyper-V version

Answer: B

MCQ 6:

How do you change the state of a VM (e.g., turn it on/off)?

A) Task Manager

B) Right-click the VM in Hyper-V Manager

C) Control Panel

D) Command Line

Answer: B

Microsoft

MCQ 7:

What is the purpose of the "Connect" option in Hyper-V?

A) Link to cloud services

B) Access the VM console

C) Enable remote desktop

D) Join a domain

Answer: B

MCQ 8:

Which Windows edition is required to enable Hyper-V?

A) Windows 10 Home

B) Windows 10 Pro/Windows Server

C) Windows 7 Ultimate

D) Windows 8.1

Answer: B

MCQ 9:

What action initiates the OS installation in a VM?

A) Allocating more CPU

B) Starting the VM from the "Connect" window

C) Enabling virtualization in BIOS

D) Creating a snapshot

Answer: B

Systems Administrator

MCQ 10:

Where is the "Turn Windows features on or off" located?

A) Device Manager

B) Programs and Features

C) Taskbar settings

D) Registry Editor

Answer: B

MCQ 11:

What can you adjust to optimize VM performance for workloads?

A) Screen saver settings

B) Memory and CPU in VM settings

C) Host OS theme

D) Monitor refresh rate

Answer: B

Microsoft

MCQ 12:

Which menu provides options to shut down or start a VM?

A) File Explorer

B) Right-click context menu in Hyper-V Manager

C) Windows Update

D) Network settings

Answer: B

MCQ 1:

What is a Virtual Machine (VM)?

A) A physical server in a data center

B) A virtual representation of a physical machine

C) A cloud storage service

D) A type of network router

Answer: B

MCQ 2:

What is the operating system running inside a VM called?

A) Host OS

B) Guest OS

C) Primary OS

D) Kernel OS

Answer: B

Systems Administrator

MCQ 3:

Which of the following is a key advantage of using VMs?

A) Increased hardware costs

B) Reduced data center costs

C) Limited scalability

D) Slower performance

Answer: B

MCQ 4:

What does the virtualization layer do?

A) Manages network traffic

B) Allocates physical resources to VMs

C) Installs the host OS

D) Provides internet connectivity

Answer: B

Microsoft

MCQ 5:

Which file contains the VM's hardware settings?

A) `.vmdk`

B) `.vmx`

C) `.vmem`

D) `vmware.log`

Answer: B

MCQ 6:

What is the purpose of a `.vmdk` file?

A) Stores VM BIOS configurations

B) Defines virtual disk characteristics

C) Logs VM activities

D) Manages snapshots

Answer: B

MCQ 7:

Which of the following is NOT a use case for VMs?

A) Desktop virtualization

B) Hardware virtualization

C) Replacing all physical hardware permanently

D) Easier migration

Answer: C

Systems Administrator

MCQ 8:

What does the `.vmsn` file store?

A) VM BIOS settings

B) Snapshot data

C) Network configurations

D) Host OS details

Answer: B

MCQ 9:

How does a VM interact with physical hardware?

A) Directly accesses hardware

B) Through the virtualization layer

C) Via cloud services

D) Using external drivers

Answer: B

Microsoft

MCQ 10:

Which feature is enabled by VMs in data centers?

A) Higher power consumption

B) Scalability (adding/upgrading VMs)

C) Limited resource allocation

D) Slower deployment times

Answer: B

MCQ 11:

What type of virtualization is provided by VMs?

A) Only network virtualization

B) Hardware and desktop virtualization

C) Storage virtualization only

D) No virtualization support

Answer: B

MCQ 12:

Which file logs VM activities?

A) `.vmx`

B) `.vmdk`

C) `vmware.log`

D) `.vmem`

Answer: C

MCQ 1:

Which of the following is a key merit of virtual machines?

A) Higher hardware costs

B) Ability to run multiple operating systems on a single host

C) Reduced portability

D) Decreased security

Answer: B

MCQ 2:

How do virtual machines help reduce costs in data centers?

A) By increasing the number of physical servers

B) By hosting multiple VMs on a single physical server

C) By eliminating the need for operating systems

D) By reducing internet bandwidth

Answer: B

MCQ 3:

What is a significant security advantage of VMs?

A) Shared user accounts across VMs

B) Isolation between VMs prevents security breaches from spreading

C) No need for firewalls

D) Automatic malware removal

Answer: B

MCQ 4:

Which of the following is a demerit of virtual machines?

A) Improved performance over physical machines

B) Lower efficiency compared to real machines

C) Higher portability

D) Reduced dependency on host hardware

Answer: B

MCQ 5:

What happens if the physical host machine fails?

A) Only one VM is affected

B) All hosted VMs become unavailable

C) VMs automatically migrate to another host

D) The host OS takes over VM operations

Answer: B

MCQ 6:

Why might VM performance be lower than a physical machine?

A) VMs use dedicated hardware resources

B) VMs share resources of the host machine

C) VMs do not use operating systems

D) VMs are immune to latency issues

Answer: B

MCQ 7:

Which of the following is a use case for VMs mentioned in the document?

A) Running only compatible software

B) Simplifying backup and disaster recovery

C) Increasing physical server requirements

D) Reducing software licensing costs

Answer: B

MCQ 8:

What is a challenge of using VMs in cloud computing?

A) Limited scalability

B) Higher operational costs

C) Inability to migrate VMs

D) Lack of isolation between VMs

Answer: B

MCQ 9:

How do VMs handle incompatible software?

A) By modifying the host OS

B) By decoupling applications from the underlying OS

C) By requiring hardware upgrades

D) By disabling security features

Answer: B

MCQ 10:

Which of the following is NOT a merit of VMs?

A) Portability

B) Cost reduction

C) Superior performance compared to physical machines

D) Security isolation

Answer: C

MCQ 11:

What is a limitation of VM efficiency?

A) Direct hardware access

B) Indirect hardware access through requests

C) Exclusive resource allocation

D) No dependency on host resources

Answer: B

MCQ 12:

Which file is associated with VM snapshots?

A) .vmx

B) .vmsn

C) vmware.log

D) .vmdk

Answer: B

MCQ 1:

What is the primary advantage of virtualization mentioned in the document?

A) Increases hardware costs

B) Transforms traditional hardware into virtual environments

C) Requires specialized physical servers

D) Limits software compatibility

Answer: B

Microsoft

MCQ 2:

How do VMs reduce data center costs?

A) By eliminating the need for operating systems

B) By hosting multiple VMs on a single physical server

C) By reducing internet bandwidth usage

D) By decreasing security requirements

Answer: B

MCQ 3:

What security benefit do VMs provide?

A) Shared user credentials across VMs

B) Isolation that prevents security breaches from affecting other VMs

C) Automatic malware detection for all VMs

D) No need for firewalls or antivirus

Answer: B

MCQ 4:

Which of these is a significant performance limitation of VMs?

A) Faster processing than physical machines

B) Shared host resources leading to lower performance

C) Inability to run multiple applications

D) Exclusive access to hardware

Answer: B

Systems Administrator

MCQ 5:

What happens when the physical host machine fails?

A) Only the primary VM is affected

B) All hosted VMs become unavailable

C) VMs automatically switch to cloud backup

D) The hypervisor creates new instances

Answer: B

MCQ 6:

Why is VM efficiency lower than physical machines?

A) Because they use dedicated hardware

B) Due to indirect hardware access through virtualization layer

C) They don't support modern operating systems

D) They have smaller memory capacity

Answer: B

Microsoft

MCQ 7:

Which operational benefit do VMs provide for administrators?

A) More complex backup procedures

B) Simplified disaster recovery and deployment

C) Increased physical server maintenance

D) Slower software testing cycles

Answer: B

MCQ 8:

What is a major cost disadvantage of using VMs in cloud computing?

A) Reduced storage requirements

B) Higher operational expenses

C) Elimination of hardware costs

D) Decreased electricity usage

Answer: B

MCQ 9:

How do VMs handle software that's incompatible with the host OS?

A) By modifying the host OS kernel

B) By running different guest operating systems

C) By disabling security features

D) By requiring hardware upgrades

Answer: B

Systems Administrator

MCQ 10:

Which of these is NOT listed as a merit of VMs?

A) Portability across environments

B) Cost reduction in data centers

C) Superior raw performance compared to physical servers

D) Ability to run multiple OS instances

Answer: C

MCQ 11:

What is the fundamental reason for VM performance limitations?

A) Dedicated hardware allocation

B) Resource sharing with other VMs on the host

C) Lack of virtualization support

D) Smaller disk space allocation

Answer: B

Microsoft

MCQ 12:

Which statement best describes VM portability?

A) VMs are permanently tied to their host hardware

B) VMs can be easily migrated between different hosts

C) VM configurations cannot be changed after creation

D) Portability decreases with more allocated resources

Answer: B

MCQ 13:

What makes VM backup procedures simpler?

A) The need to back up each physical server individually

B) The ability to capture entire VM states as single files

C) Automatic cloud synchronization features

D) Built-in RAID configurations

Answer: B

MCQ 14:

Why might cloud-based VMs be more expensive?

A) They eliminate hardware costs completely

B) They involve ongoing service fees and resource charges

C) They don't require any management

D) They use less electricity than physical servers

Answer: B

Systems Administrator

MCQ 15:

What critical infrastructure component represents a single point of failure for VMs?

A) Individual VM operating systems

B) The physical host machine

C) Network switches

D) Cooling systems

Answer: B

MCQ 1:

What is the primary purpose of virtual networking?

A) To replace all physical network hardware permanently

B) To create software-based versions of networking components

C) To increase hardware dependency in data centers

D) To reduce network flexibility

Answer: B

MCQ 2:

Which of the following is NOT a virtual networking component?

A) Virtual switch

B) Virtual router

C) Physical Ethernet cable

D) Virtual firewall

Answer: C

MCQ 3:

How does virtual networking improve network agility?

A) By requiring more physical hardware

B) By enabling software-based modifications without hardware changes

C) By limiting network configurations

D) By reducing scalability

Answer: B

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MCQ 4:

What advantage does virtual networking provide for data centers?

A) Increases hardware costs

B) Allows stretching across multiple physical locations

C) Requires manual reconfiguration of physical devices

D) Decreases network management efficiency

Answer: B

Systems Administrator

MCQ 5:

Why is virtual networking beneficial for network administrators?

A) It complicates network modifications

B) It simplifies network management and configuration changes

C) It requires purchasing new hardware for every change

D) It reduces the ability to provision networks

Answer: B

MCQ 6:

Which statement best describes the flexibility of virtual networking?

A) It locks networks into fixed configurations

B) It allows networks to be provisioned according to specific needs

C) It depends entirely on physical hardware limitations

D) It slows down deployment times

Answer: B

Systems Administrator

MCQ 7:

What is eliminated by using virtual networking solutions?

A) The need for IP addresses

B) The requirement to frequently buy or replace physical hardware

C) All security risks

D) Network administrators

Answer: B

MCQ 8:

Which task becomes easier with virtual networking?

A) Physical server maintenance

B) Modifying network configurations

C) Manufacturing network hardware

D) Installing fiber optic cables

Answer: B

MCQ 9:

What does virtual networking enable in multi-location data centers?

A) Isolated networks per location

B) Unified management across locations

C) Increased hardware redundancy

D) Slower data transfer speeds

Answer: B

MCQ 10:

Which of these is a key characteristic of virtual networking?

A) Hardware-dependent operations

B) Software-defined network management

C) Permanent network configurations

D) Limited scalability

Answer: BMCQ 1:

What is the primary function of a virtual switch?

A) To replace physical routers

B) To enable communication between VMs and the physical network

C) To provide VPN services

D) To monitor network traffic

Answer: B

MCQ 2:

Which virtual networking tool implements security policies between subnets?

A) Virtual switch

B) Virtual router

C) Virtual load balancer

D) Virtual network adapter

Answer: B

MCQ 3:

What is the role of virtual firewalls?

A) To balance network traffic

B) To enforce security policies in virtual networks

C) To replace physical cabling

D) To connect remote locations

Answer: B

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MCQ 4:

Which tool improves performance by distributing traffic across multiple VMs?

A) Virtual VPN

B) Virtual load balancer

C) Virtual switch

D) Virtual security group

Answer: B

MCQ 5:

What does a virtual VPN provide?

A) Local storage for VMs

B) Secure remote access to virtual networks

C) Hardware redundancy

D) Physical network segmentation

Answer: B

MCQ 6:

How do virtual network adapters function?

A) By replacing physical servers

B) By connecting VMs to virtual networks

C) By monitoring network performance

D) By enforcing routing policies

Answer: B

MCQ 7:

What is the purpose of virtual network security groups?

A) To allocate CPU resources

B) To define access controls for network traffic

C) To create physical network partitions

D) To replace virtual routers

Answer: B

MCQ 8:

Which tool provides visibility into virtual network performance?

A) Virtual firewall

B) Virtual network monitoring tools

C) Virtual load balancer

D) Virtual switch

Answer: B

Microsoft

MCQ 9:

Which of these is NOT a virtual networking tool?

A) Virtual router

B) Virtual load balancer

C) Physical Ethernet hub

D) Virtual firewall

Answer: C

MCQ 10:

What advantage do virtual networking tools offer over physical ones?

A) Higher hardware costs

B) Software-based flexibility and scalability

C) Limited configuration options

D) Slower deployment times

Answer: B

MCQ 11:

Which tool would you use to segment traffic between virtual subnets?

A) Virtual load balancer

B) Virtual router

C) Virtual network adapter

D) Virtual monitoring tool

Answer: B

Systems Administrator

MCQ 12:

What is a key feature of virtual load balancers?

A) Encrypting all network traffic

B) Distributing traffic to enhance performance and availability

C) Replacing physical firewalls

D) Creating virtual cables

Answer: B

Summary of Key Tools:

1. **Virtual Switch**: VM communication
2. **Virtual Router**: Subnet routing + security
3. **Virtual Firewall**: Policy enforcement
4. **Virtual Load Balancer**: Traffic distribution
5. **Virtual VPN**: Secure remote access
6. **Virtual Security Groups**: Traffic filtering
7. **Monitoring Tools**: Performance visibility

MCQ 1:

What is the primary function of a virtual switch?

A) To replace physical routers

B) To provide network connectivity to virtual machines

C) To encrypt all network traffic

D) To monitor CPU usage

Answer: B

MCQ 2:

Which type of virtual switch allows VMs to communicate with the physical network?

A) Private vSwitch

B) External vSwitch

C) Internal vSwitch

D) Hybrid vSwitch

Answer: B

MCQ 3:

What is the purpose of an Internal vSwitch?

A) To connect VMs to the internet

B) To enable communication between VMs and the Hyper-V host

C) To replace physical firewalls

D) To balance network traffic

Answer: B

MCQ 4:

Which vSwitch type is completely isolated from the physical network?

A) External vSwitch

B) Private vSwitch

C) Internal vSwitch

D) Public vSwitch

Answer: B

MCQ 5:

When would you use a Private vSwitch?

A) For VMs needing internet access

B) For isolated VM communication on the same host

C) To connect to external storage

D) To replace physical switches

Answer: B

MCQ 6:

How does an External vSwitch differ from an Internal vSwitch?

A) It cannot connect to VMs

B) It provides physical network access to VMs

C) It only works with Linux VMs

D) It requires no configuration

Answer: B

MCQ 7:

Which statement about Internal vSwitches is TRUE?

A) They replace physical routers

B) They allow host-VM communication

C) They provide internet access

D) They are used for load balancing

Answer: B

MCQ 8:

What is a key advantage of virtual switches over physical ones?

A) Higher hardware costs

B) Software-based management and flexibility

C) Limited scalability

D) Slower performance

Answer: B

MCQ 9:

Which vSwitch type would you use for a VM needing internet access?

A) Private vSwitch

B) External vSwitch

C) Internal vSwitch

D) Local vSwitch

Answer: B

MCQ 10:

What is common to all vSwitch types?

A) They require physical cables

B) They connect virtual machines

C) They provide VPN services

D) They replace network adapters

Answer: B

MCQ 11:

Which vSwitch is ideal for secure testing environments?

A) External vSwitch

B) Private vSwitch

C) Internal vSwitch

D) Public vSwitch

Answer: B

MCQ 12:

What limitation does a Private vSwitch have?

A) No VM-to-VM communication

B) No physical network access

C) Incompatibility with Hyper-V

D) Slower speeds than physical switches

Answer: B

Summary of vSwitch Types:

1. **External**: Physical network access for VMs
2. **Internal**: Host-VM communication
3. **Private**: Isolated VM-to-VM communication

MCQs on Standard Switch Architecture

MCQ 1:

What is the primary function of a standard switch's I/O interfaces?

A) To power the switch components

B) To serve as ports where data enters and exits the switch

C) To store the switch's configuration

D) To manage the operating system

Answer: B

MCQ 2:

Which component of a standard switch is responsible for forwarding packets between I/O interfaces?

A) Processor

B) Switch Fabric

C) Memory

D) Power supply

Answer: B

MCQ 3:

What role does the processor play in a standard switch?

A) Provides electrical power

B) Manages switch configuration and packet forwarding decisions

C) Acts as a physical port for data transfer

D) Replaces the need for memory

Answer: B

MCQ 4:

Where is the switch's configuration and forwarding information stored?

A) Switch Fabric

B) Memory

C) Power supply

D) I/O interfaces

Answer: B

MCQ 5:

Which component ensures the switch receives power?

A) Operating system

B) Power supply

C) Processor

D) Management interface

Answer: B

Systems Administrator

MCQ 6:

What is the purpose of the operating system in a standard switch?

A) To physically connect cables

B) To run the switch's software

C) To replace the switch fabric

D) To act as a management interface

Answer: B

MCQ 7:

Which interface is used to configure and manage the switch?

A) I/O interfaces

B) Management interface (CLI/GUI)

C) Switch Fabric

D) Memory

Answer: B

MCQ 8:

What is the switch fabric analogous to in a standard switch?

A) The switch's power source

B) The internal pathway for packet forwarding

C) The external ports

D) The configuration storage

Answer: B

MCQ 9:

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Which of these is NOT a component of standard switch architecture?

A) Memory

B) Processor

C) Virtual firewall

D) Power supply

Answer: C

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MCQ 10:

How does the management interface typically interact with users?

A) Via physical cables

B) Through CLI or GUI

C) By automating all configurations

D) Using wireless signals

Answer: B

Systems Administrator

Summary of Key Components:

1. **I/O Interfaces**: Data entry/exit points.
2. **Switch Fabric**: Internal packet forwarding.
3. **Processor**: Configuration and decision-making.
4. **Memory**: Stores configuration/data.

5. **Power Supply**: Provides electricity.
6. **OS**: Runs switch software.
7. **Management Interface**: CLI/GUI for configuration.

MCQs on Network Security and Compliance

MCQ 1:

What is the primary purpose of network policies?

A) To increase network congestion

B) To govern network use and protect data/devices

C) To replace security measures

D) To limit bandwidth for all applications

Answer: B

CERTIFIED

MCQ 2:

Which of the following is NOT a network security measure?

A) Firewalls

B) Encryption

C) Traffic shaping

D) Access controls

Answer: C *(Traffic shaping manages bandwidth, not security)*

MCQ 3:

What does traffic shaping ensure?

A) Equal bandwidth for all applications

B) Priority for critical applications and congestion prevention

C) Elimination of network policies

D) Automatic hardware replacement

Answer: B

Microsoft®

MCQ 4:

What is NIC teaming used for?

A) Reducing the number of network interfaces

B) Combining interfaces for bandwidth, failover, and load balancing

C) Disabling security protocols

D) Blocking unauthorized access

Answer: B

MCQ 5:

Which component provides automatic backup during outages?

A) Traffic shaping

B) Failover

C) Network policies

D) Encryption

Answer: B

MCQ 6:

What do firewalls and intrusion detection systems protect against?

A) Hardware failures

B) Unauthorized access and security threats

C) Bandwidth limitations

D) NIC teaming errors

Answer: B

Systems Administrator

MCQ 7:

Which technique prevents network congestion?

A) Failover

B) Traffic shaping

C) NIC teaming

D) Access controls

Answer: B

MCQ 8:

What is a key benefit of NIC teaming?

A) Reduced security

B) Increased bandwidth and redundancy

C) Slower data transfer

D) Manual failover

Answer: B

MCQ 9:

Which aspect ensures continuity during system failures?

A) Network policies

B) Failover

C) Traffic shaping

D) Encryption

Answer: B

MCQ 10:

What role do access controls play in security?

A) Manage traffic flow

B) Restrict unauthorized access

C) Combine network interfaces

D) Replace firewalls

Answer: B

Summary of Key Concepts:

1. **Network Policies**: Rules for governance/protection.
2. **Security**: Firewalls, encryption, access controls.
3. **Traffic Shaping**: Bandwidth management.
4. **NIC Teaming**: Combines interfaces for performance.
5. **Failover**: Automatic backup during outages.

MCQs on Virtual Storage

MCQ 1:

What is the primary function of virtual storage?

A) To replace all physical storage permanently

B) To create a virtual representation of storage devices

C) To eliminate the need for operating systems

D) To reduce internet bandwidth usage

Answer: B

MCQ 2:

How does virtual storage improve resource utilization?

A) By limiting storage capacity

B) By pooling physical storage and allocating it dynamically

C) By disabling data backups

D) By requiring manual partitioning

Answer: B

Systems Administrator

MCQ 3:

Which feature of virtual storage enhances data protection?

A) Traffic shaping

B) Snapshots and replication

C) NIC teaming

D) Failover

Answer: B

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MCQ 4:

What is a key benefit of virtual storage in data management?

A) Increased hardware complexity

B) Simplified provisioning and allocation

C) Slower performance

D) Fixed storage sizes

Answer: B

MCQ 5:

How does virtual storage support scalability?

A) By locking storage capacity

B) By allowing seamless addition of storage resources

C) By reducing data availability

D) By eliminating backups

Answer: B

MCQ 6:

What flexibility does virtual storage provide?

A) Single storage configuration for all applications

B) Customizable virtual devices with unique characteristics

C) Manual resource allocation

D) No disaster recovery options

Answer: B

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MCQ 7:

Which performance improvement does virtual storage offer?

A) Slower data access

B) Caching and data compression

C) Increased hardware dependency

D) Limited storage pools

Answer: B

Systems Administrator

MCQ 8:

How does virtual storage aid in disaster recovery?

A) By preventing data replication

B) By enabling quick relocation of virtual storage devices

C) By eliminating snapshots

D) By reducing storage capacity

Answer: B

MCQ 9:

What abstracts physical storage in virtual storage systems?

A) Network policies

B) A software layer

C) Hardware controllers

D) Power supplies

Answer: B

Systems Administrator

MCQ 10:

Which task becomes easier with virtual storage?

A) Physical hardware maintenance

B) Resizing storage devices

C) Manual data migration

D) Network congestion management

Answer: B

Summary of Virtual Storage Benefits:

1. **Abstraction**: Virtual representation of physical storage.
2. **Efficiency**: Dynamic allocation and pooling.
3. **Protection**: Snapshots, replication, backups.
4. **Scalability**: Seamless capacity expansion.
5. **Performance**: Caching, compression, de-duplication.
6. **Disaster Recovery**: Quick relocation/cloud storage.

MCQs on Types of Data Stores

MCQ 1:

Which type of database stores data in tables with rows and columns?

A) NoSQL

B) Relational

C) Key-value

D) Time-series

Answer: B

MCQ 2:

What is the primary query language used in relational databases?

A) NoQL

B) SQL

C) Python

D) XML

Answer: B

MCQ 3:

Which database type is optimized for handling unstructured/semi-structured data?

A) Relational

B) NoSQL

C) Columnar

D) File-based

Answer: B

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MCQ 4:

Which of these is an example of a NoSQL database?

A) MySQL

B) MongoDB

C) Oracle

D) SQL Server

Answer: B

MCQ 5:

What type of database stores data as objects?

A) Relational

B) Object-oriented

C) Time-series

D) Columnar

Answer: B

MCQ 6:

Which database is best suited for time-stamped data?

A) Key-value

B) Time-series

C) File-based

D) Relational

Answer: B

MCQ 7:

What is a key feature of columnar databases?

A) Slow query performance

B) Faster retrieval for analytical queries

C) Only supports structured data

D) Uses SQL exclusively

Answer: B

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MCQ 8:

Which database stores data as key-value pairs?

A) MongoDB

B) Redis

C) MySQL

D) InfluxDB

Answer: B

MCQ 9:

Which of these is a file-based database example?

A) Cassandra

B) JSON files

C) Oracle

D) CouchDB

Answer: B

MCQ 10:

What distinguishes NoSQL from relational databases?

A) Uses only SQL

B) Handles unstructured data flexibly

C) Requires fixed schemas

D) Slower for large datasets

Answer: B

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Summary of Database Types:

1. **Relational (SQL)**: Tables, rows, columns (MySQL, Oracle).
2. **NoSQL**: Unstructured data (MongoDB, Cassandra).
3. **Object-oriented**: Data as objects (MongoDB).
4. **Time-series**: Time-stamped data (InfluxDB).
5. **Columnar**: Column-based storage (Cassandra).
6. **Key-value**: Unique key pairs (Redis).
7. **File-based**: JSON/XML files.

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MCQs on Full Virtualization

MCQ 1:

What is the primary characteristic of full virtualization?

A) Guest OS is modified to work with the hypervisor

B) Guest OS runs unmodified and unaware of virtualization

C) Only supports single VM per host

D) Requires no hypervisor

Answer: B

MCQ 2:

Which layer manages virtualization in full virtualization?

A) Guest OS

B) Hypervisor (VMM)

C) Physical hardware

D) Network switch

Answer: B

MCQ 3:

What executes application/user-level instructions directly in full virtualization?

A) Hypervisor

B) Underlying hardware

C) Guest OS kernel

D) Virtual switch

Answer: B

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MCQ 4:

Which of these is a full virtualization hypervisor example?

A) Docker

B) VMware ESXi

C) Kubernetes

D) LXC

Answer: B

MCQ 5:

What is a key disadvantage of full virtualization?

A) Guest OS requires modifications

B) Performance overhead due to binary translation

C) Limited to one VM per host

D) No hardware independence

Answer: B

MCQ 6:

How does a hypervisor interact with the guest OS in full virtualization?

A) Modifies the guest OS kernel

B) Translates OS-level requests at runtime

C) Shares memory with the guest OS

D) Bypasses hardware entirely

Answer: B

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MCQ 7:

What happens if the physical server fails in full virtualization?

A) Only one VM is affected

B) All VMs on the host are impacted

C) VMs auto-migrate to another host

D) Hypervisor continues running

Answer: B

MCQ 8:

Which privilege level does the hypervisor typically operate at?

A) Ring 3 (User mode)

B) Ring 1 (Higher privilege than guest OS)

C) Ring 0 (Kernel mode)

D) Ring 2 (Unused)

Answer: B *(Hypervisor runs at Ring 1, guest OS at Ring 2/3, hardware at Ring 0)*

MCQ 9:

What type of instructions are directly executed by hardware in full virtualization?

A) OS kernel calls

B) Application/user-level instructions

C) Hypervisor management commands

D) Network packet routing

Answer: B

MCQ 10:

Which virtualization technique is contrasted with full virtualization?

A) Hardware-assisted virtualization

B) Paravirtualization (requires guest OS modifications)

C) Containerization

D) Network virtualization

Answer: B

Summary of Key Concepts:

1. **Definition**: Guest OS runs unmodified on hypervisor.
2. **Hypervisor Role**: Manages hardware access (Ring 1).
3. **Performance**: Overhead from binary translation.
4. **Disadvantage**: Single point of failure (host crash affects all VMs).
5. **Examples**: VMware ESXi, VirtualBox.

MCQs on Para Virtualization

MCQ 1:

What is the primary difference between para virtualization and full virtualization?

A) Para virtualization does not use a hypervisor

B) Guest OS is modified in para virtualization

C) Full virtualization requires modified guest OS

D) Hardware is emulated in para virtualization

Answer: B

MCQ 2:

Which component handles hyper calls in para virtualization?

A) Guest OS kernel

B) Hypervisor

C) Physical hardware

D) Virtual switch

Answer: B

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MCQ 3:

What replaces hardware calls in para virtualization?

A) Binary translation

B) Hyper calls

C) System interrupts

D) Network packets

Answer: B

MCQ 4:

Which of these is a key advantage of para virtualization?

A) Supports unmodified Windows OS

B) Reduced virtualization overhead

C) Complete hardware isolation

D) Slower performance

Answer: B

MCQ 5:

Why is para virtualization less portable?

A) It doesn't use a hypervisor

B) Requires guest OS modifications

C) Only works with proprietary hardware

D) Lacks hardware abstraction

Answer: B

MCQ 6:

Which hypervisor is commonly associated with para virtualization?

A) VMware ESXi

B) Xen

C) VirtualBox

D) Hyper-V

Answer: B

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MCQ 7:

What is a limitation of para virtualization?

A) High hardware emulation overhead

B) Limited to open-source OSs like Linux

C) Cannot run multiple VMs

D) Hypervisor operates at Ring 3

Answer: B

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MCQ 8:

How does para virtualization improve performance?

A) By emulating all hardware

B) By using hyper calls instead of binary translation

C) By disabling the hypervisor

D) By isolating hardware completely

Answer: B

MCQ 9:

What is required for a guest OS to work in para virtualization?

A) No changes

B) Special drivers/modifications

C) Physical server reboot

D) Dedicated GPU

Answer: B

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MCQ 10:

Which statement is true about para virtualization?

A) Guest OS is unaware of virtualization

B) Hardware resources are shared, not emulated

C) Uses binary translation extensively

D) Supports all proprietary OSs without modification

Answer: B

Summary of Key Concepts:

1. **Definition**: Guest OS is modified to collaborate with hypervisor.
2. **Hyper Calls**: Replace hardware calls for efficiency.
3. **Advantages**: Lower overhead, better performance.
4. **Disadvantages**: OS modification required, limited compatibility.
5. **Example**: Xen hypervisor.

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MCQs on Full Virtualization vs Para Virtualization

MCQ 1:

What is the key difference in Guest OS handling between Full and Para Virtualization?

A) Full Virtualization requires OS modifications

B) Para Virtualization requires a modified Guest OS

C) Full Virtualization doesn't use a hypervisor

D) Para Virtualization emulates all hardware

Answer: B

MCQ 2:

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How does the Guest OS interact with hardware in Full Virtualization?

A) Via hyper calls to the hypervisor

B) Through direct hardware calls

C) Using shared memory

D) By bypassing the hypervisor

Answer: B

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MCQ 3:

Which technique uses hyper calls for hardware access?

A) Full Virtualization

B) Para Virtualization

C) Both

D) Neither

Answer: B

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MCQ 4:

Why is Full Virtualization more portable?

A) It requires specialized hardware

B) It supports unmodified Guest OSs

C) It lacks a hypervisor

D) It uses binary translation exclusively

Answer: B

MCQ 5:

Which virtualization method offers better performance?

A) Full Virtualization

B) Para Virtualization

C) Both are equal

D) Depends on hardware

Answer: B

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MCQ 6:

What is a security advantage of Para Virtualization?

A) Guest OS is unaware of virtualization

B) Hypervisor mediates all hardware access

C) No hypervisor is needed

D) Uses unmodified OS kernels

Answer: B

MCQ 7:

Which statement is true about Full Virtualization?

A) Guest OS knows it's virtualized

B) Uses binary translation for hardware calls

C) Hypervisor is optional

D) Only works with Linux

Answer: B

MCQ 8:

What limits Para Virtualization's compatibility?

A) Lack of hypervisor support

B) Need for Guest OS modifications

C) Inability to run multiple VMs

D) High hardware costs

Answer: B

MCQ 9:

Which hypervisor is commonly used in Para Virtualization?

A) VMware ESXi

B) Xen

C) VirtualBox

D) Hyper-V

Answer: B

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MCQ 10:

What is emulated in Full Virtualization but shared in Para Virtualization?

A) CPU registers

B) Hardware resources

C) Network bandwidth

D) Storage controllers

Answer: B

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Summary of Key Differences:

Feature	Full Virtualization	Para Virtualization
Guest OS	Unmodified	Modified
Hardware Access	Hardware calls → Binary translation	Hyper calls → Direct access

Feature	Full Virtualization	Para Virtualization
Performance	Lower (overhead)	Higher
Portability	High (supports any OS)	Low (needs OS support)
Security	Less secure	More secure
Example Hypervisor	VMware ESXi, VirtualBox	Xen

MCQs on Containers

MCQ 1:

What is a container in software terms?

A) A full virtual machine with its own OS

B) A lightweight unit packaging code, libraries, and dependencies

C) A physical server component

D) A type of hypervisor

Answer: B

MCQ 2:

How do containers differ from virtual machines (VMs)?

A) Containers include a full OS

B) Containers share the host OS kernel

C) VMs are lighter than containers

D) Containers require hardware emulation

Answer: B

MCQ 3:

Which component allows containers to access the OS kernel?

A) Hypervisor

B) Container engine

C) Virtual switch

D) BIOS

Answer: B

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MCQ 4:

Why are containers considered lightweight?

A) They emulate hardware

B) They omit the guest OS layer

C) They run only on cloud platforms

D) They use binary translation

Answer: B

MCQ 5:

What makes containers portable?

A) Dependency on host OS configurations

B) Self-contained dependencies in each image

C) Requirement of specialized hardware

D) Use of hyper calls

Answer: B

MCQ 6:

Which of these is a containerization platform?

A) VMware ESXi

B) Docker

C) Xen

D) Hyper-V

Answer: B

MCQ 7:

Where do containers sit in the system architecture?

A) Below the hardware layer

B) On top of the host OS

C) Between hypervisor and VM

D) Replaces the kernel

Answer: B

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MCQ 8:

What is virtualized in containerization?

A) Hardware resources

B) The OS kernel (namespace isolation)

C) Network switches

D) Storage devices

Answer: B

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MCQ 9:

Which statement is true about containers?

A) Slower to start than VMs

B) Ideal for microservices and cloud apps

C) Require guest OS installations

D) Cannot run multiple instances on one host

Answer: B

MCQ 10:

Which cloud service supports containers?

A) AWS EC2

B) Amazon ECS

C) VMware vSphere

D) Oracle VirtualBox

Answer: B

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Summary of Key Concepts:

1. **Definition**: Lightweight, OS-level virtualization.
2. **Components**: Container engine, host OS, shared kernel.
3. **Advantages**: Portability, speed, resource efficiency.
4. **Use Cases**: Microservices, cloud-native apps.
5. **Platforms**: Docker, AWS ECS, Azure, GCP.

MCQs on Merits and Demerits of Containers

MCQ 1:

Why are containers considered lightweight?

A) They include a full OS

B) Their images are small (MBs vs GBs)

C) They require dedicated hardware

D) They use binary translation

Answer: B

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MCQ 2:

What is a key advantage of containers over VMs?

A) Higher resource overhead

B) Faster boot times

C) Requires guest OS installations

D) Slower deployment

Answer: B

MCQ 3:

How does container isolation improve security?

A) By sharing the host OS kernel

B) By running applications in self-contained environments

C) By eliminating dependencies

D) By using hypervisors

Answer: B

MCQ 4:

What enables container portability?

A) Dependency on host OS configurations

B) Encapsulation of all dependencies

C) Use of hardware emulation

D) Large image sizes

Answer: B

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MCQ 5:

Which type of applications face challenges in containerization?

A) Stateless microservices

B) Stateful applications (e.g., databases)

C) Web servers

D) CLI tools

Answer: B

MCQ 6:

What is a security risk in containerized environments?

A) Dedicated kernels per container

B) Shared host OS kernel vulnerabilities

C) Lack of isolation

D) Slow boot times

Answer: B

MCQ 7:

Why do containers have less overhead than VMs?

A) They emulate hardware

B) They share the host OS kernel

C) They require full OS installations

D) They use hypervisors

Answer: B

MCQ 8:

What is a limitation of container dependency management?

A) No dependency packaging needed

B) Complex dependency bundling

C) Automatic dependency resolution

D) Eliminates runtime environments

Answer: B

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MCQ 9:

Which operational efficiency is highlighted for containers?

A) Slower scaling

B) Faster deployment than VMs

C) Higher resource consumption

D) Requires physical servers

Answer: B

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MCQ 10:

What makes containers less suitable for legacy systems?

A) Lightweight nature

B) Dependency on modern kernel features

C) Fast boot times

D) Small image sizes

Answer: B

Summary of Key Points:

Aspect	Merits	Demerits
Size	Lightweight (MBs)	Dependency packaging complexity
Performance	Fast boot, low overhead	Shared kernel security risks
Portability	"Write once, run anywhere"	Stateful app challenges
Security	Isolation benefits	Cross-container vulnerabilities

MCQs on Virtualization vs Containerization

MCQ 1:

What is the primary difference between virtualization and containerization?

A) Virtualization uses containers, while containerization uses VMs

B) Virtualization virtualizes hardware, while containerization virtualizes the OS

C) Both require a hypervisor

D) Containers include a full OS

Answer: B

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MCQ 2:

Which technology uses a hypervisor to create VMs?

A) Containerization

B) Virtualization

C) Both

D) Neither

Answer: B

MCQ 3:

What is the key advantage of containerization over virtualization?

A) Higher hardware utilization

B) Faster boot times and lower overhead

C) Better isolation between applications

D) Supports legacy OSes

Answer: B

MCQ 4:

Which component manages containers?

A) Hypervisor

B) Container Engine

C) BIOS

D) Virtual Switch

Answer: B

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MCQ 5:

Why do VMs have slower boot times than containers?

A) They lack dependencies

B) They require a full OS to load

C) They use less memory

D) They don't virtualize hardware

Answer: B

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MCQ 6:

What cost does containerization primarily reduce?

A) Hardware costs

B) Software deployment costs

C) Electricity consumption

D) Network bandwidth

Answer: B

MCQ 7:

Which statement is true about VMs?

A) They share the host OS kernel

B) Each VM includes its own OS

C) They are lighter than containers

D) They boot faster than containers

Answer: B

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MCQ 8:

What is abstracted in containerization?

A) Physical hardware

B) The host OS kernel

C) Network switches

D) Storage devices

Answer: B

MCQ 9:

Which technology is better for running isolated, multi-OS environments?

A) Virtualization

B) Containerization

C) Both equally

D) Neither

Answer: A

MCQ 10:

What makes containers more portable than VMs?

A) Larger image sizes

B) Self-contained dependencies

C) Hardware emulation

D) Slower deployment

Answer: B

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Summary of Key Differences:

Feature	Virtualization	Containerization
Abstraction Layer	Hardware	OS Kernel
Component	Hypervisor + VMs	Container Engine + Containers
Boot Time	Slower (full OS)	Faster (shared kernel)
Use Case	Multi-OS environments	Microservices, cloud-native apps
Cost Efficiency	Reduces hardware costs	Reduces software deployment costs

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